

Ruihan Henry Liu

PH.D. CANDIDATE · UNIVERSITY OF CALIFORNIA, BERKELEY

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Education

University of California, Berkeley

DOCTOR OF PHILOSOPHY (PH.D.) IN PHYSICS

Advisors: Uroš Seljak and Simone Ferraro

MASTER OF ARTS (MA) IN PHYSICS

Berkeley, California

Expected May 2027

Dec. 2024

University of British Columbia

B.SC. (HONOURS) IN PHYSICS AND MATHEMATICS

Thesis: *On Bayesian Baseline Removal and Signal Extraction Methods in Astronomy*

Advisor: Douglas Scott

Vancouver, B.C., Canada

May 2021

Publications and Preprints

First Author Publications:

Model-Independent Measurement of Baryon Gas Fractions through Galaxy-Galaxy Lensing and the Kinematic Sunyaev-Zel'dovich Effect

Manuscript In Prep

LIU, R. H., SELJAK, U., FERRARO, S., HADZHIYSKA, B., *et al.*

Baryons Beyond Halos: Unbound Gas, Stars, and the Sunyaev-Zel'dovich Effect

Manuscript In Prep

LIU, R. H., FERRARO, S., HADZHIYSKA, B.

Fast Baryonic Field Painting for Sunyaev-Zel'dovich Analyses: Transfer Function vs. Hybrid Effective Field Theory

Phys. Rev. D 113, 063558

LIU, R. H., HADZHIYSKA, B., FERRARO, S., BOSE, S., HERNÁNDEZ-AGUAYO, C.

arXiv: 2504.11794

Measurements of the Thermal Sunyaev-Zel'dovich Effect with ACT and DESI Luminous Red Galaxies

Phys. Rev. D 112, 083561

LIU, R. H., FERRARO, S., SCHAAN, E., *et al.*

arXiv: 2502.08850

Coupling parsec and gigaparsec scales: primordial non-Gaussianity with multi-tracer intensity mapping

Phys. Rev. D 103, 063520

LIU, R. H. AND BREYSSE, P. C.

arXiv: 2002.10483

A machine learning approach for identifying the counterparts of submillimetre galaxies and applications to the GOODS-North field

MNRAS, 489, 2, 1770-1786

LIU, R. H., HILL, R., SCOTT, D., ALMAINI, O., AN, F., GUBBELS, C., HSU, L., LIN, L., SMAI, I., STACH, S.

arXiv: 1901.09594

Other Publications:

Detection of the pairwise kinematic Sunyaev-Zel'dovich effect and pairwise velocity with DESI DR1 galaxies and ACT DR6 and Planck CMB data

Phys. Rev. D 113, 063538

GONG, Y., GALLARDO, P. A., BEAN, R., MOORE, J., VAVAGIAKIS, E. M., ..., LIU, R. H., ..., *et al.*

arXiv: 2511.23417

Evidence for large baryonic feedback at low and intermediate redshifts from kinematic Sunyaev-Zel'dovich observations with ACT and DESI photometric galaxies

Phys. Rev. D 112, 083509

HADZHIYSKA, B., FERRARO, S., RIED GUACHALLA, B., SCHAAN, E., ..., LIU, R. H., ..., *et al.*

arXiv: 2407.07152

Talks and Presentations

Lawrence Berkeley National Laboratory CMB Lunch

Berkeley, California

MODEL-INDEPENDENT MEASUREMENTS OF BARYON FEEDBACK WITH KSZ AND LENSING

Apr. 2026

New Synergies in Multi-Probe Cosmology

KITP, Santa Barbara, California

PROBING BOUND AND UNBOUND GAS WITHIN SIMULATIONS TO UNDERSTAND THE SUNYAEV-ZEL'DOVICH EFFECTS
(POSTER)

Feb. 2026

CAMELS Collaboration Telecon

Virtual

FAST RECONSTRUCTION OF BARYONIC SIMULATION FIELDS USING CAMELS AND MILLENNIUMTNG

Jan. 2026

DESI Collaboration Winter Meeting

Tucson, Arizona

PROBING BOUND AND UNBOUND GAS WITH THE SUNYAEV-ZEL'DOVICH EFFECTS

Dec. 2025

Stanford Cosmology Seminar

Stanford, California

PROBING BARYONIC MATTER IN THE UNIVERSE WITH THE SUNYAEV-ZEL'DOVICH EFFECT

Mar. 2025

Lawrence Berkeley National Laboratory CMB Lunch

Berkeley, California

PROBING BARYONIC MATTER IN THE UNIVERSE WITH THE SUNYAEV-ZEL'DOVICH EFFECT

Jan. 2025

Cosmology in the Adriatic

Split, Croatia

COMPARISON OF LAGRANGIAN AND EULERIAN EFT METHODS FOR BARYON PASTING ON SIMULATIONS

Jul. 2024

DESI Collaboration Summer Meeting

Marseille, France

DESI X ACT: CROSS CORRELATION FOR DETECTION OF THE THERMAL SUNYAEV-ZEL'DOVICH EFFECT

Jul. 2024

Awards and Honours

Outstanding Graduate Student Instructor Award, UC Berkeley

2025

NSERC Postgraduate Scholarship – Doctoral (PGS-D), Natural Sciences and Engineering Research Council of Canada

2022 - 2025

Sushant Singh Rajput Memorial Fellowship, UC Berkeley

2021

NSERC Undergraduate Student Research Award, Natural Sciences and Engineering Research Council of Canada

2018, 2019

Trek Excellence Scholarship, UBC

2017 - 2020

Awarded to students in the top 5% of their faculty and year. Received during every eligible term.

Various Other Undergraduate Scholarships, UBC

2016 - 2021

Teaching Experience

Department of Physics, UC Berkeley

Berkeley, California

GRADUATE STUDENT INSTRUCTOR

2021 - present

- Taught courses on Electromagnetism and Optics, Bayesian Statistics and Machine Learning, and Introductory Physics.
- Received the Outstanding Graduate Student Instructor Award for teaching.

Department of Physics and Astronomy, University of British Columbia

Vancouver, British Columbia

UNDERGRADUATE TEACHING ASSISTANT

Sept. 2018 - Apr 2019

- PHYS 101: Energy and Waves (taught for two terms).
- Received the Undergraduate Teaching Excellence Award from the UBC Science Undergraduate Society.

Volunteering, Service and Outreach

Astronomy On Tap East Bay

Oakland, California

SPEAKER

2026

- Public Speaker for Astronomy on Tap East Bay

UC Berkeley Astro Night

Berkeley, California

VOLUNTEER

2024 - 2025

- Volunteer with UC Berkeley Astro Night responsible for astronomy outreach with the public.

UofT AstroTours

Toronto, Ontario

VOLUNTEER

Jun. 2019 - Aug. 2019

- AstroTours is an outreach event held monthly by the University of Toronto Department of Astronomy and Astrophysics for the general public.